

Bahati Patrick

Data Scientist | Machine Learning | Big Data Engineering

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PROFILE SUMMARY

Data Scientist with hands-on experience in machine learning, deep learning, and big data engineering, turning complex datasets into actionable insights and predictive models. Currently completing a **Master's Degree in Data Science** at the **Università degli Studi di Messina (Italy)**, with applied research experience in spiking neural networks and computational neuroscience. Comfortable communicating findings to both technical and non-technical audiences.

EDUCATION

Master's Degree in Data Science 2024 – 2026
Università degli Studi di Messina *Messina, Italy*

Bachelor's Degree in Business Information Technology 2020 – 2023
Nkumba University *Entebbe, Uganda* **CGPA: 4.45 / 5.00**

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

AI & Business Automation 2025 – 2026
Academic Rapido TechFloor *Funded by the Accenture Foundation*

- Built AI-powered business automation workflows using n8n, OpenAI, and REST/Google Workspace APIs for enterprise process optimization.
- Applied prompt engineering, RAG pipelines, and multi-agent workflow design across real-world automation projects.

Next.js App Router Fundamentals August 2025
Vercel *Certificate of Completion*

- Completed hands-on training in Next.js App Router architecture, server components, routing, data fetching, and full-stack deployment on Vercel.

WORK EXPERIENCE

Computational Neuroscience Research Intern March 2026 – June 2026
Università degli Studi di Messina — Brian Lab *Messina, Italy*

- Implemented biologically inspired Spiking Neural Networks (SNNs) using the Brian2 simulator.
- Developed and simulated Leaky Integrate-and-Fire (LIF) and conductance-based neuron models.
- Designed Spike-Timing-Dependent Plasticity (STDP) and Reward-Modulated STDP (R-STDP) learning systems.
- Investigated excitatory/inhibitory synaptic dynamics and eligibility trace mechanisms.
- Visualized membrane voltage, synaptic conductance, and adaptive weight changes including LTP and LTD.
- Built end-to-end experimental SNN simulation pipelines for computational neuroscience research.

Business Coach October 2023 – October 2024
AVSI Foundation *Isingiro District, Nakivale, Uganda*

- Mentored micro and small enterprises using data-informed business growth strategies and market analysis.
- Collaborated with private sector employers on job placement, collecting and reporting labor market data for workforce development programs.

Data Analyst Intern June 2022 – September 2022
Stratcom Communications and Computer Solutions *Uganda*

- Cleaned, processed, and analyzed datasets to support data-driven business decisions.
- Developed dashboards and automated reporting solutions using Python and BI tools.
- Supported system performance monitoring and optimization through real-time data analysis.

Data Clerk & Research Assistant 2021 – 2022
Windle International Uganda *Uganda*

- Managed large-scale field data collection, validation, and database maintenance for education-focused research.
- Performed quantitative and qualitative analysis, ensuring data accuracy and compliance with ethical research standards.

- Facilitated communication between development stakeholders and local communities, delivering capacity-building programs and monitoring project outcomes.

SELECTED PROJECTS

Reward-Modulated Spiking Neural Network (R-STDP) *Brian2, Python*

- Designed a full conductance-based LIF neural network with excitatory and inhibitory synapse populations.
- Implemented STDP and reward-modulated synaptic plasticity (R-STDP) with eligibility traces.
- Simulated spike-timing-dependent learning dynamics and visualized LTP/LTD weight evolution.
- Produced publication-quality figures of membrane voltage, firing rates, and synaptic conductance.

Plant Disease Detection Using CNN *TensorFlow, Keras, Python*

- Built a CNN-based deep learning model for multi-class plant disease classification using TensorFlow/Keras.
- Trained on the PlantVillage dataset: 70,000+ images across 38 disease classes.
- Applied data augmentation, dropout regularization, batch normalization, and early stopping.
- Achieved ~91% test accuracy on held-out images.

Real-Time Credit Card Fraud Detection Pipeline *PySpark, Kafka, Hadoop, Docker, Streamlit*

- Architected a **Lambda Architecture** pipeline processing **280,000+ transactions**, combining Spark Structured Streaming and Kafka (**sub-2-second latency**) with Hadoop MapReduce batch analytics.
- Trained distributed ML models (Random Forest, Logistic Regression) via Spark MLlib on imbalanced datasets, delivering results through a Streamlit/Plotly dashboard containerized with Docker Compose.

Big Data Analytics Final Project — Apache HBase & FDA CAERS Dataset

Hadoop, HBase, Docker, Python

University Project

- Designed and deployed a multi-node Hadoop/HBase cluster (HDFS, YARN, ZooKeeper) using Docker to store and query real-world FDA adverse-event data.
- Built a Python ETL pipeline converting **149,945** raw JSON adverse-event reports (**233,821** linked product records) into HBase-ready TSV files, bulk-loaded via MapReduce ImportTsv.
- Designed a normalized HBase schema (namespaces, column families, versioning, compression) and implemented DDL, CRUD/DML, and cluster administration operations via the HBase Shell.

AI Customer Support Automation Platform *n8n, OpenAI, Supabase, Airtable, Zendesk*

GitHub

- Designed an AI-powered customer support workflow integrating semantic search, automated ticket classification, and Zendesk escalation.
- Built a Retrieval-Augmented Generation (RAG) chatbot using a Supabase Vector Store, with automated Airtable logging and HTML/CSS analytics dashboards.

Multi-Agent Marketing Campaign Generator *n8n, Google Gemini, Google Docs*

GitHub

- Developed a multi-agent AI system with specialized agents for campaign planning, content generation, and quality review.
- Automated brand strategy and editorial plan generation with consistent tone enforcement via Google Docs integration.

TECHNICAL SKILLS

Programming	Python, SQL, Java, JavaScript, HTML/CSS
Machine Learning	Scikit-learn, TensorFlow, Keras, PyTorch, Spiking Neural Networks (Brian2), STDP, R-STDP
Scientific Computing	NumPy, Pandas, SciPy, Simulation Modeling, Feature Engineering
Big Data	Apache Spark (PySpark, Spark SQL), Apache Kafka, Hadoop (HDFS, MapReduce), Apache HBase, Apache Hive
Databases	MySQL, PostgreSQL, MongoDB, SQL Server, Oracle
Data Visualization	Matplotlib, Seaborn, Plotly, Streamlit, Power BI, Tableau, Excel
Computer Vision	OpenCV, CNNs for Image Classification
Web & Backend	React, Angular, Node.js, Next.js (App Router), Spring Boot, ASP.NET Core, Django
AI & Automation	n8n, OpenAI, Google Gemini, RAG, Multi-Agent Workflows, Prompt Engineering, REST APIs
DevOps & Tools	Docker, Docker Compose, Kubernetes, Git, GitHub
Research Methods	Data Preprocessing, Statistical Analysis, Academic Writing, Experimental Design

LANGUAGES

English	Full Professional Proficiency
Italian	Intermediate
Kinyarwanda	Full Professional Proficiency
Swahili	Intermediate
Luganda	Full Professional Proficiency